

# THE MITOCHONDRIAL RESPIRATORY CHAIN AND OXIDATIVE PHOSPHORYLATION

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Course for first-year medical students  
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## **Course objectives:**

- 1- Define the concepts of a redox couple and redox potential
- 2- Establish the relationship between the redox potential of a reaction and Gibbs free energy
- 3- Outline the mechanism of oxidative phosphorylation
- 4- Understand the relationship between electron transfer by respiratory chain complexes and ATP synthesis by ATP synthase.
- 5- Explain the mechanisms of inhibition and uncoupling of the respiratory chain.

## **Course outline:**

- I. Introduction
- II. Redox potential and free energy change
- III. The mitochondrial respiratory chain (MRC)
  - III.1. Mitochondria: organelles with high metabolic activity
  - III.2. Organization of the MRC
  - III.3. The sequence of electron transport in the mitochondrial membrane
  - III.4. Complex V: ATP synthase
- IV. Chemiosmotic hypothesis: P. Mitchell (1961)
- V. Energy balance:
- VI. Regulation of MCR
- VII. Mitochondrial respiratory chain dysfunction

## I. INTRODUCTION

The energy stored in carbohydrates, lipids, and proteins must be converted into ATP (rapidly usable energy), which is produced in the mitochondria from MCR and oxidative phosphorylation.

The respiratory chain and oxidative phosphorylation are the final stages of energy catabolism. They enable the production of ATP from reduced coenzymes (NADH, FADH<sub>2</sub>) resulting from the breakdown of nutrients. This ATP is then used in anabolic reactions. Located in the inner membrane of the mitochondria, these reactions are essential for cell survival. They illustrate the close link between catabolism, anabolism, and energy production.

## II. REDOX POTENTIAL AND FREE ENERGY CHANGE

Most biochemical reactions are redox reactions.

An oxidation-reduction reaction is a reaction in which two chemical compounds exchange electrons. One of the compounds will give up electrons (electron-rich) and the other will receive them (electron-deficient).

We define:

- **Oxidation:** reaction involving the loss of  $e^-$  by a reducing agent (red).  

$$\text{Reducing agent 1} \longrightarrow \text{Oxidizing agent 1} + ne^-$$
- **Reduction:** reaction involving the gain of electrons ( $e^-$ ) by an oxidizing agent (ox).  

$$\text{Oxidizing agent 2} + ne^- \longrightarrow \text{Reducing agent 2}$$
- **Oxidation-reduction reaction:**

The sum of the two reactions above (half reactions) occurs simultaneously.

This involves a transfer of  $e^-$  between a red and an ox belonging to two different redox couples (red1/ox1) and (red2/ox2).

- **Reducing equivalent:** Corresponds to an  $e^-$  transferred in a redox reaction.
- **Oxidizing agent:** A chemical species (atom, ion, molecule, etc.) capable of capturing one or more electrons.
- **Reducing agent:** A chemical species (atom, ion, molecule, etc.) capable of donating one or more electrons.
- **Ox/red pair:** Refers to a reducing agent and its conjugate oxidizing agent.

For each redox couple, an oxidation-reduction potential is defined, which provides information on the ability of the oxidant to gain electrons, or of the reducing agent to lose them (electron transfer energy).

- **Definition of redox potential**

Electromotive force defined by the attractive force of electrons created by the difference in affinity for these electrons between two chemical compounds, one an electron donor and the other an electron acceptor.

The redox potential  $E$  is expressed as a potential difference (unit: Volt V) at a standard couple ( $H^+/H_2$ ).

### - Standard redox potential $E^\circ$ :

We work under standard conditions:

- T: 25°C
- Pressure: 1 atm
- Concentrations of oxidized and reduced forms: 1 M

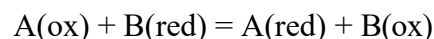
In biochemistry, the pH is set at 7, and the redox potential is denoted as  $E'^\circ$ .

If two redox couples are brought together, the couple with the lowest redox potential will oxidize, and the couple with the highest redox potential will reduce.

- The most reducing compounds have the lowest redox potential. They are called electronegative.
- The most oxidizing compounds have the highest redox potential. They are called electropositive.

### • Relationship between redox potential and free energy change:

Oxidation-reduction reactions are accompanied by a loss of free energy, just like any other spontaneous reaction. The standard change in free energy during a reaction of the type:



Can be calculated from the standard redox potentials of the two couples involved in the reaction using the equation:

$$\Delta G^\circ = -nF\Delta E^\circ$$

$n$ : number of electrons transferred.

$F$ : Faraday's constant (23.063 kcal/V · mol or KJ/V), (96493.5 Coulombs.equivalent<sup>-1</sup>)

$E^\circ$ : V;

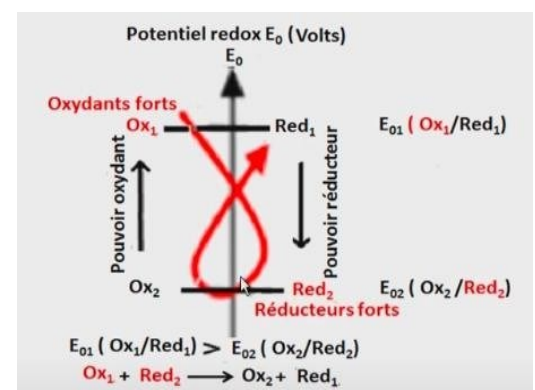
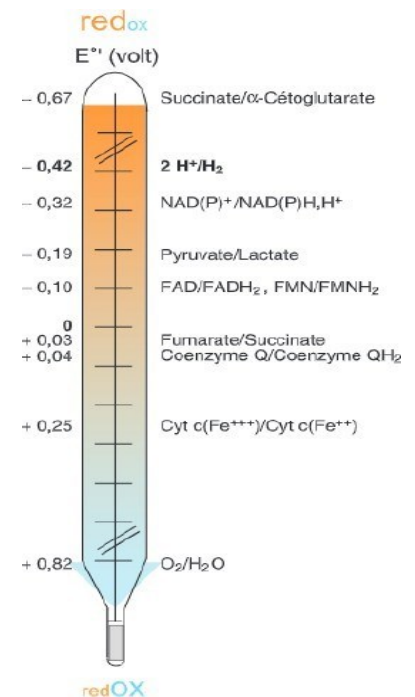
$\Delta G^\circ$ : kcal/mole

### Note:

- The direction of electron transfer in a redox reaction can be predicted from the redox potential. Using the gamma rule:

\* The higher the redox potential of a couple ( $E'^\circ_1$ ), the stronger the oxidizing agent in the couple and the weaker the reducing agent.

\* Conversely, the lower the redox potential ( $E'^\circ_2$ ), the stronger the reducing agent in the couple and the weaker its oxidizing agent.



\*Therefore: any redox couple oxidizes another couple with a redox potential lower than its own potential ( $E^{\circ}1 > E^{\circ}2$ ;  $\Delta E^{\circ} > 0$ ).

- For a redox reaction to occur spontaneously, the following conditions must be met:  $\Delta G^{\circ} < 0$ ;  $\Delta E^{\circ} > 0$

| Équation à l'électrode                                                                                                    | $E^{\circ}(\text{V})$ |
|---------------------------------------------------------------------------------------------------------------------------|-----------------------|
| Succinate + $\text{CO}_2 + 2\text{H}^+ + 2\text{e}^- \rightleftharpoons \alpha\text{-cétoglutarate} + \text{H}_2\text{O}$ | -0,670                |
| Acétate + $2\text{H}^+ + 2\text{e}^- \rightleftharpoons$ acétaldéhyde                                                     | -0,580                |
| $2\text{H}^+ + 2\text{e}^- \rightleftharpoons \text{H}_2$                                                                 | -0,421                |
| $\alpha\text{-Cétoglutarate} + \text{CO}_2 + 2\text{H}^+ + 2\text{e}^- \rightleftharpoons$ isocitrate                     | -0,380                |
| Cystine + $2\text{H}^+ + 2\text{e}^- \rightleftharpoons$ 2 cystéines                                                      | -0,340                |
| $\text{NAD}^+ + 2\text{H}^+ + 2\text{e}^- \rightleftharpoons \text{NADH} + \text{H}^+$                                    | -0,320                |
| $\text{NADP}^+ + 2\text{H}^+ + 2\text{e}^- \rightleftharpoons \text{NADPH} + \text{H}^+$                                  | -0,324                |
| Acétaldéhyde + $2\text{H}^+ + 2\text{e}^- \rightleftharpoons$ éthanol                                                     | -0,197                |
| Pyruvate + $2\text{H}^+ + 2\text{e}^- \rightleftharpoons$ lactate                                                         | -0,185                |
| Oxaloacétate + $2\text{H}^+ + 2\text{e}^- \rightleftharpoons$ malate                                                      | -0,166                |
| $\text{FAD} + 2\text{H}^+ + 2\text{e}^- \rightleftharpoons \text{FADH}_2$ (dans les flavoprotéines)                       | +0,031                |
| Fumarate + $2\text{H}^+ + 2\text{e}^- \rightleftharpoons$ succinate                                                       | +0,031                |
| Ubiquinone + $2\text{H}^+ + 2\text{e}^- \rightleftharpoons$ ubiquinol                                                     | +0,045                |
| $2 \text{ cytochrome } b_{(\text{ox})} + 2\text{e}^- \rightleftharpoons 2 \text{ cytochrome } b_{(\text{réd})}$           | +0,070                |
| $2 \text{ cytochrome } c_{(\text{ox})} + 2\text{e}^- \rightleftharpoons 2 \text{ cytochrome } c_{(\text{réd})}$           | +0,254                |
| $2 \text{ cytochrome } a_{3(\text{ox})} + 2\text{e}^- \rightleftharpoons 2 \text{ cytochrome } a_{3(\text{réd})}$         | +0,385                |
| $\frac{1}{2} \text{O}_2 + 2\text{H}^+ + 2\text{e}^- \rightleftharpoons \text{H}_2\text{O}$                                | +0,816                |

The greater the potential difference between two couples, the more the reaction progresses under standard conditions towards the products before reaching a state of equilibrium.

Let's look at the oxidation reaction of NADH, a powerful reducing agent, by a molecule of oxygen  $\text{O}_2$ , a powerful oxidizing agent.



The standard redox potentials of the two couples can be written as follows:



The voltage change for the overall reaction is equal to the difference between the two  $E^{\circ}$  values ( $\Delta E^{\circ}$ ):

$$\Delta E^{\circ} = +0.82 \text{ V} - (-0.32 \text{ V}) = +1.14 \text{ V}$$

which represents an estimate of the free energy released when NADH is oxidized by molecular oxygen under standard conditions. By introducing this value into the previous equation,

$$\begin{aligned} \Delta G^{\circ} &= (-2) (23.063 \text{ kcal/V} \cdot \text{mol}) (1.14 \text{ V}) \\ &= -52.6 \text{ kcal/mol of oxidized NADH} \end{aligned}$$

The energy released during the oxidation of NADH by  $\text{O}_2$  is sufficient to produce several ATP molecules: ( $\Delta G^{\circ} = +7.3 \text{ kcal/mol}$ ).

The transfer of this energy from NADH to ATP in the mitochondria occurs through a series of short steps that release energy.

Electrons are transferred to NAD<sup>+</sup> (or FAD) in the mitochondria from several substrates in the TCA cycle: isocitrate, alpha-ketoglutarate, malate, and succinate. Among these intermediates, the first three have relatively high negative redox potentials, high enough for electron transfer to NAD<sup>+</sup>.

On the other hand, the oxidation of succinate to fumarate, which has a more positive redox potential, is carried out by the reduction of FAD, a coenzyme with a higher affinity for electrons than NAD<sup>+</sup>.

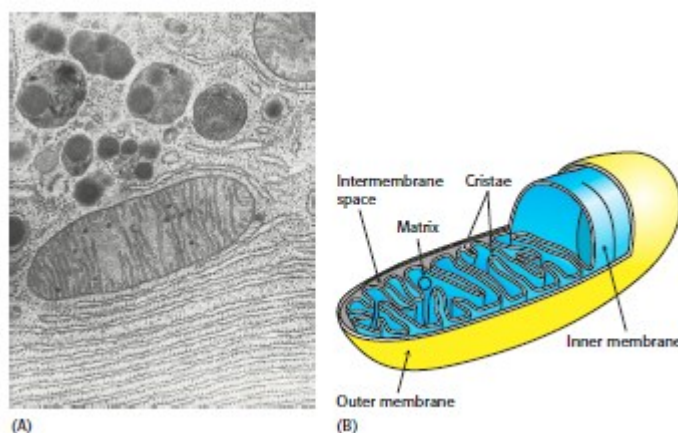
### III. THE MITOCHONDRIAL RESPIRATORY CHAIN (MRC)

This is the set of biochemical mechanisms through which the energy released during electron transfer in the respiratory chain is used for the phosphorylation of ADP to ATP.

#### III.1 The mitochondrion: an organelle with high metabolic activity

It is an organelle, ellipsoidal in shape. It comprises two membranes and two internal compartments, the matrix and the intermembrane space.

- **The outer membrane:** this membrane is highly permeable thanks to porins, channels that allow molecules smaller than 10 kDa to pass through.
- **The intermembrane space** is a volume that can increase when mitochondrial respiration is intense. Its pH is 0.75 units lower than that of the matrix.
- **The inner membrane** forms numerous folds, or cristae, which greatly increase the membrane surface area. Permeable to water and respiratory gases (O<sub>2</sub>, CO<sub>2</sub>), it is completely impermeable to protons (H<sup>+</sup>) and carries a whole range of specialized transporters and shuttles that control the transfer of substances (ATP, Pi, ADP, pyruvate, fatty acids) between the matrix and the cytosol. It is the site of the respiratory chain and oxidative phosphorylation.
- **The matrix** This compartment is the site of numerous oxidative reactions (Krebs cycle,  $\beta$  oxidation) that ensure the production of reducing power (NADH, H<sup>+</sup>, FADH<sub>2</sub>).



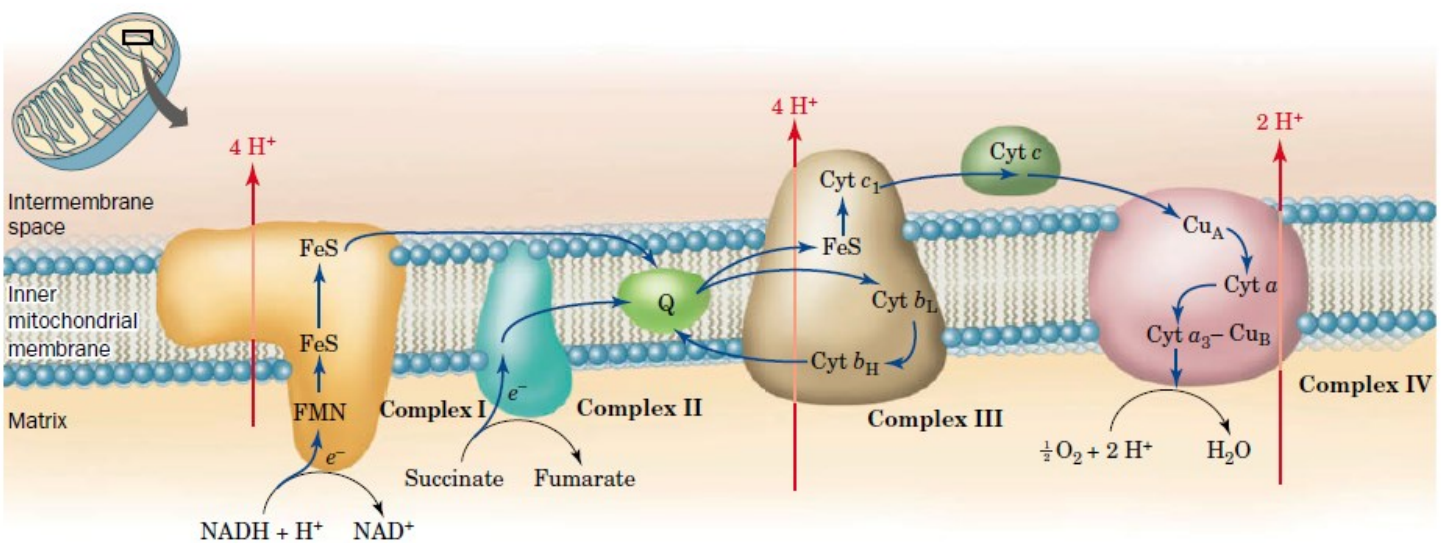
### III.2 Organization of the MCR

The electron transport chain consists mainly of proteins. The redox properties of the chain are due to **prosthetic groups** associated with **oxidoreductase** enzymes embedded in the inner mitochondrial membrane.

The prosthetic groups responsible for electron transfer are: **iron-sulfur** centers, copper-copper centers, **flavins** in the form of **FMN** (flavin mononucleotide) or **FAD** (flavin adenine dinucleotide), **cytochromes**, and **ubiquinone** (known as coenzyme Q).

Some proteins in the chain have proton transporter (pump) properties. The respiratory chain is made up of:

- Four large, immobile complexes labeled: **I, II, III, IV**,
- and two mobile elements: **Cytochrome c** and **Ubiquinone (UQ)**.



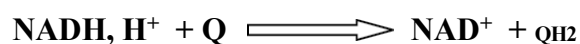
#### III.2.1- Fixed elements

##### ➤ **Complex I: NADH-CoQ oxidoreductase (NADH dehydrogenase)**

It is a large L-shaped transmembrane protein. Containing:

- NADH dehydrogenase with FMN (a prosthetic group; flavin mononucleotide)
- Several centers (Fe-S).

It catalyzes the transfer of hydrogen ( $2e^-$ ,  $2H^+$ ) from NADH,  $H^+$  ( $E^\circ = -0.32 \text{ V}$ ) to ubiquinone ( $E^\circ = +0.045 \text{ V}$ ), translocating electrons via FMN and iron-sulfur centers, according to the following overall reaction:



This reaction is highly exergonic ( $\Delta G^\circ = -51 \text{ kJ/mol}$ ) and allows the translocation of four protons from the matrix to the intermembrane space. Complex I is a proton pumping site.

NB: NADH, H<sup>+</sup> has a dual origin:

- Mitochondrial:  $\beta$ -oxidation of fatty acids, oxidative decarboxylation of pyruvate to acetyl CoA, and the Krebs cycle.
- Cytosolic: glycolysis (malate-aspartate shuttle).
- **Complex II (succinate-CoQ oxidoreductase):**

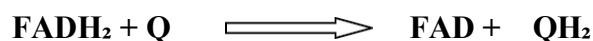
Located on the matrix side of the inner mitochondrial membrane. Contains:

- The enzyme succinate dehydrogenase with coenzyme FAD (an enzyme catalyzes the sixth reaction of the citric acid cycle).
- Several centers (Fe-S).

This complex establishes a direct link between the Krebs cycle and the respiratory chain. It receives the reducing equivalents of FADH<sub>2</sub> produced by the citric acid cycle and passes them to coenzyme Q via the center (Fe-S) proteins.

This complex also receives FADH<sub>2</sub> formed during  $\beta$ -oxidation, as well as FADH<sub>2</sub> from the glycerol-3-phosphate shuttle (NADH, H<sup>+</sup> of glycolytic origin).

According to the reaction:



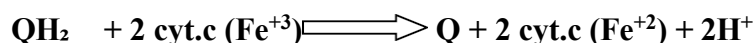
At this stage, the change in standard Gibbs free energy, corresponding to  $\Delta G^\circ = -2.9 \text{ kJ/mol}$ , is not exergonic enough to allow proton transfer. As a result, less ATP is formed by the oxidation of FADH<sub>2</sub> than by NADH, H<sup>+</sup>.

- **Complex III (CoQ-cytochrome c- Oxidoreductase or cytochrome bc1 complex):**

Complex III contains:

- 2 cytochromes b (b<sub>562</sub> and b<sub>566</sub>)
- Cytochrome c<sub>1</sub>
- A center protein (Fe-S)

It receives the reduced equivalents (e-reduced) of CoQH<sub>2</sub> and passes them to cytochrome c via cytochromes b and c<sub>1</sub> and the center protein (Fe-S). According to:



The standard free energy change for this step,  $\Delta G^\circ = -37 \text{ kJ/mol}$ , is exergonic and allows the translocation of four protons from the matrix to the intermembrane space.

Complex III is another proton pump.

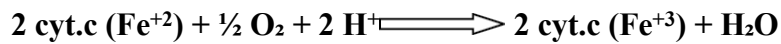
- **Complex IV (cytochrome c oxidase):**

This complex contains:

- 2 cytochromes a (a and a<sub>3</sub>).
- 2 Cu proteins.

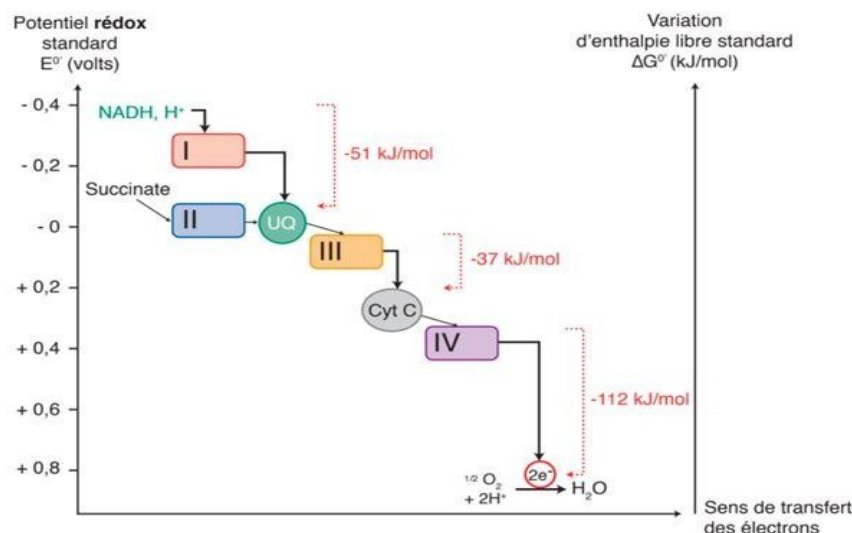


It receives the reducing equivalents from cytochrome C and passes them on to molecular oxygen via cytochromes a and a<sub>3</sub>. According to:



The water thus produced is the terminal electron acceptor and the final product of the respiratory chain. The standard free energy change for this step,  $\Delta G^\circ = -112 \text{ kJ}\cdot\text{mol}^{-1}$ , is exergonic and allows the translocation of two protons.

Therefore, complex IV acts as a proton pump.



### III.2.2- Mobile elements

#### ➤ Coenzyme Q or Ubiquinone

A small hydrophobic molecule.

Enables the transition between complex I or II and complex III thanks to its mobility within the inner mitochondrial membrane.

Present in excess compared to other constituents.

Ubiquinone Q, receiving reducing equivalents from complex I or II, is reduced to ubiquinol:  $\text{QH}_2$ . It transports all reducing equivalents from oxidative catabolism.

#### ➤ 2- Cytochrome c

Water-soluble hemoprotein, mobile on the outer surface of the inner mitochondrial membrane. Enables the transition between complex III and complex IV.

### III.3 The sequence of electron transport in the mitochondrial membrane

The free energy required for the synthesis of ATP from ADP and  $\text{P}_i$  is generated in stages by the oxidation of NADH,  $\text{H}^+$ , and  $\text{FADH}_2$  in the MCR.

Through this chain, electrons move from a low standard reduction potential  $E^\circ$  to a high standard reduction potential. In other words, they are transferred in the direction of the redox potential gradient from the most negative to the most positive.



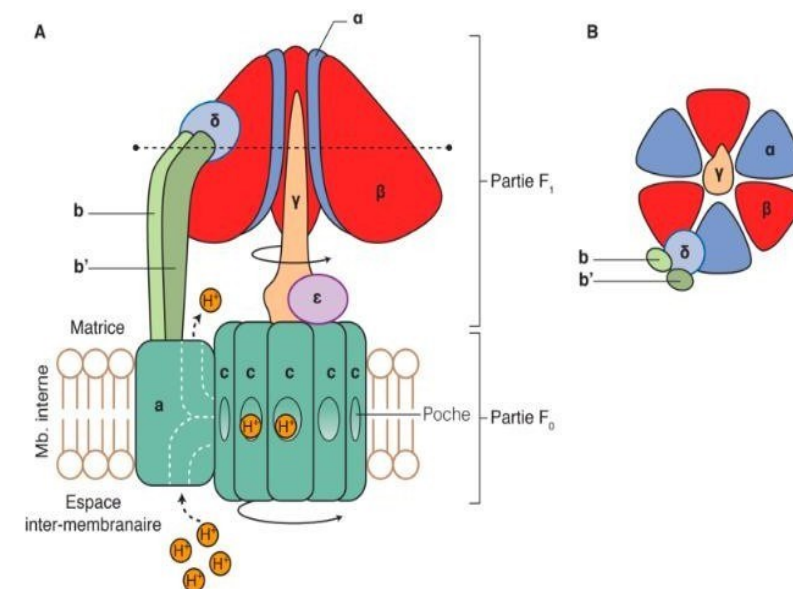
From complexes I and II, electrons are transported to complex III by coenzyme Q (CoQ or ubiquinone). They are then transported to complex IV by the membrane protein cytochrome c. Complex IV carries out the final stage of electron transfer to oxygen.

### III.4 Complex V: ATP Synthase

ATP synthase is a reverse ion pump which, instead of transporting protons in the opposite direction to the concentration gradient, causes ATP synthesis through the passage of protons in the direction of the gradient. ATP synthase is called a pedunculated sphere and consists of two parts, F<sub>0</sub> and F<sub>1</sub>:

- **The F<sub>0</sub> part** is membrane-bound. It consists of four distinct polypeptides that form a transmembrane channel that transports protons from the intermembrane space to the matrix.
- **Part F<sub>1</sub>** protrudes from the membrane and is immersed in the mitochondrial matrix. It is composed of several subunits.
  - The  $\beta$  subunits have a catalytic site responsible for the phosphorylation of ADP to ATP.
  - The  $\alpha$  subunits have allosteric regulatory sites for ATP and ADP, which allow F<sub>1</sub> to modulate the rate of ATP production according to the availability of substrates and products.

ATP synthase couples the production of one ATP to the passage of three to four protons.



A. Reconstitution longitudinale partielle et détail des parties F<sub>1</sub> ( $\alpha$ ,  $\beta$ ,  $\gamma$ ,  $\delta$ ,  $\epsilon$ ) et F<sub>0</sub> (a, b, b', c)  
 B. Coupe transversale de la partie F<sub>1</sub> au niveau du trait de coupe.  
 Les sous-unités ( $\gamma$ ) et (c) sont des pièces mobiles dont le sens de rotation est indiqué.

## IV. CHEMIOSMOTIC HYPOTHESIS: P. MITCHELL (1961)

She assumes that e-transfer and ATP synthesis were coupled by a proton gradient that formed across the inner mitochondrial membrane.

- **Conversion of chemical energy into osmotic energy**

The energy released during oxidation (chemical energy) allows complexes I, III, and IV to export protons from the mitochondrial matrix (less acidic) to the intermembrane space (more acidic), transporting them against the electrochemical gradient.

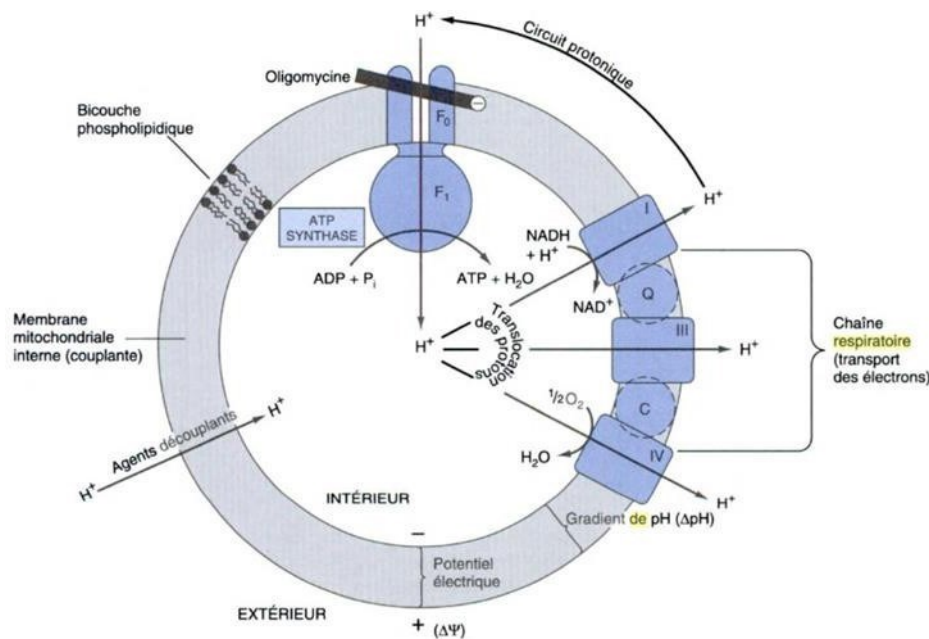
A concentration gradient of  $H^+$  is established on either side of the inner membrane, and the energy is in osmotic form.

- **Chemo-osmotic coupling**

Complex V (mitochondrial ATPase), on the other hand, allows  $H^+$  ions to return from the intermembrane space to the matrix and uses the energy produced to phosphorylate ADP into ATP, transferring the osmotic energy of the gradient in chemical form:

- For NADH, 10 protons are pumped: Synthesis of 3 ATP molecules.

For  $FADH_2$ , 6 protons are pumped: Synthesis of 2 ATP molecules. The ATP produced by this process comes from an osmo-chemical coupling.

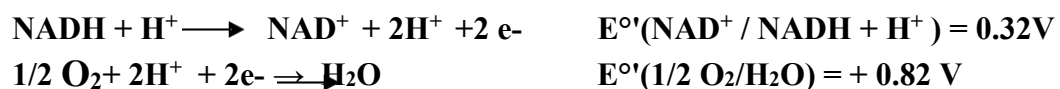


## V. ENERGY BALANCE:

The high-potential transfer electrons of NADH,  $H^+$  and  $FADH_2$  are ultimately transferred to molecular oxygen  $O_2$ , which is reduced to  $H_2O$ . In the series of redox reactions that take place during the transfer of electrons throughout the respiratory chain, if we consider only:

- The first proton and electron donor ( $NADH + H^+$ )
- The final electron acceptor (molecular oxygen,  $O_2$ )

❖ The two redox half-reactions are:



❖ Overall reaction:



$$\Delta E^{\circ'} = E^{\circ'}(1/2 O_2 / H_2O) - E^{\circ'}(NAD^+ / NADH + H^+) = + 1.14 V$$

$$\Delta G^{\circ'} = - 220 \text{ kJ/mole } (- 52.6 \text{ kcal/mole})$$

ATP is produced according to this reaction:



Theoretically, the reoxidation of one NADH,  $\text{H}^+$  allows the synthesis of 7 ATP. We calculate the energy production yield:  $(3/7) * 100 = 42.8\%$ . The rest of the energy is dissipated as heat.

- **Energy balance of glycolysis and shuttle system**

The inner mitochondrial membrane is impermeable to the reduced coenzyme NADH,  $\text{H}^+$  produced at the cytosolic level. This coenzyme uses two shuttle systems to transfer its reducing power ( $2 e^-$ ) to the mitochondrial matrix.

a- The malate-oxaloacetate shuttle (in the heart, liver and kidneys)

This shuttle regenerates NADH,  $\text{H}^+$  in the mitochondrial matrix. Therefore, 3 ATPs are formed from a cytosolic NADH,  $\text{H}^+$ .

b- The glycerol-3-phosphate shuttle (in muscles and brain)

The cytosolic NADH,  $\text{H}^+$  is regenerated in the matrix in the form of FADH. Therefore, only 2 ATP will be produced from a cytosolic NADH,  $\text{H}^+$ .

### Overall assessment

The choice of the shuttle determines the total efficiency of the oxidation of a glucose molecule:

- 38 ATP if the malate-aspartate shuttle is used.
- 36 ATP if the ...

## VI. REGULATION OF MCR

The human body needs energy equivalent to the hydrolysis of 200 moles of ATP every day, but at any given moment, the body contains less than 0.1 moles of ATP. There must therefore be a continuous flow of biosynthesis and consumption.

ATP production is regulated to meet the metabolic needs of tissues.

- **Substrate concentration**

The respiratory chain requires a sufficient amount of oxygen, NADH,  $\text{H}^+$ , and ADP. CRM is stimulated if the **NAD<sup>+</sup>/NADH**, **H<sup>+</sup>**, and **ATP/ADP**, **P<sub>i</sub>** ratios are low.

One of the limiting factors is the speed at which ADP and  $\text{P}_i$  are transported from the cytoplasm to the mitochondrial matrix by ATP/ADP translocase.

- **UCP protein**

Protons dissipate into the matrix outside of ATP synthase through a decoupling mechanism involving a protein called thermogenin (Uncoupling Protein – UCP). This protein acts as a proton channel, and this dissipation is not coupled with ATP formation but releases heat.

This natural decoupling is activated naturally by the sympathetic nervous system and thyroid hormones.

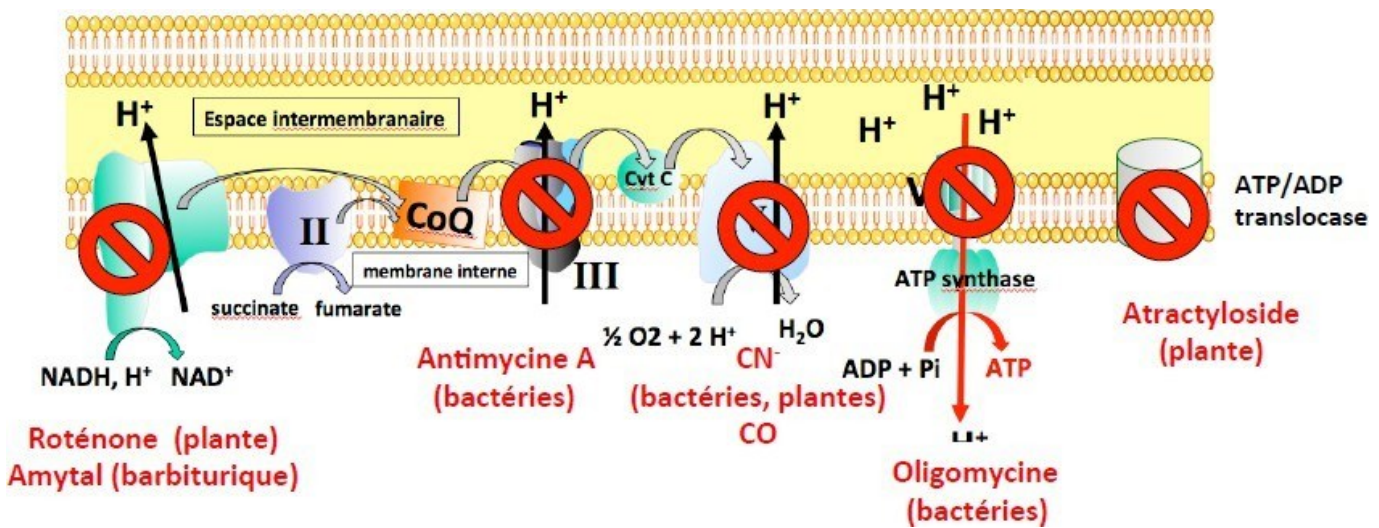
## VII. ABNORMALITIES IN THE FUNCTIONING OF THE MITOCHONDRIAL RESPIRATORY CHAIN

### 1/ Respiratory chain inhibitors and uncoupling agents:

These are highly toxic substances, and inhibition can occur at different levels: Complex I, III, IV, V, and ATP/ADP translocase. The interaction with the inhibitor is based on the similarity of the inhibitor's structure to a coenzyme or the complex's substrate.

Four categories of inhibitors are known, grouped according to their mechanisms of action.

| Agent class                                    | Cible principale                       | mechanism of action                                                                                                                                                                                                                               | Exemples                                                                                                                                                                                                                                                    |
|------------------------------------------------|----------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Electron transport inhibitors</b>           | Respiratory chain (complexes I to IV)  | Blocking the transfer of electrons at specific points in the chain results in the following cascade:<br>- No reduction of dioxygen<br>- Absence of proton pumping<br>- No creation of electrochemical gradient<br>- Stopping the synthesis of ATP | * <b>Complex I:</b> Rotenone, Piericidin A, Amytal<br>* <b>Complex II:</b> Thenoyltrifluoroacetone<br>* <b>Complex III:</b> Antimycin A, Dimercaprol<br>* <b>Complex IV:</b> Cyanide (HCN), H <sub>2</sub> S, Azide (N <sub>3</sub> ), Carbon monoxide (CO) |
| <b>Inhibitors of oxidative phosphorylation</b> | ATP synthase (portion F <sub>0</sub> ) | Blockage of the ATP synthase proton channel, and arrest of ADP phosphorylation.<br>Finally, a counterpressure is exerted on the respiratory chain leading to the secondary stop of electron transport.                                            | <b>Oligomycine,</b>                                                                                                                                                                                                                                         |
| <b>Uncouplers</b>                              | inner mitochondrial membrane           | H leakage independently of ATP synthase. Therefore:<br>Stop of ATP synthesis<br>Intense stimulation of respiration.<br>- Dissipation of energy in the form of heat.                                                                               | <b>2,4-dinitrophenol (DNP), Dicoumarol</b>                                                                                                                                                                                                                  |
| <b>Ionophores</b>                              | inner mitochondrial membrane           | Transport or channeling of ions across the membrane, with indirect inhibition of ATP synthesis.                                                                                                                                                   | <b>Valinomycine,</b>                                                                                                                                                                                                                                        |



## 2/ Genetic deficiencies: mitochondrial cytopathies:

Mitochondrial cytopathies can be defined as hereditary abnormalities in the functioning of the MCR, with ATP deficiency as their common pathophysiological mechanism.

In addition to energy deficiency, oxidative stress and an imbalance between metabolites also contribute to the development of the disease. Depending on the genetic origin of these pathologies, they can be divided into two types:

- Mitochondrial cytopathies caused by a mutation in mtDNA
- Mitochondrial cytopathies caused by a mutation in the nuclear DNA

The first clinical manifestations of these diseases are neurological, cardiac, and muscular, as a result of these organs' dependence on mitochondrial ATP synthesis to maintain their functions.

Examples of mitochondrial cytopathies: Leigh syndrome, hereditary Leber optic neuropathy (LHON), MELAS syndrome (mitochondrial encephalomyopathy) and MERRF syndrome (myoclonic epilepsy).

## 4/ Thyroid disorders:

One of the major functions of thyroid hormones is the regulation of energy metabolism. Thyroxine stimulates the synthesis of several MCR proteins, including cytochrome c oxidase, ATP synthase, and ATP/ADP translocase, resulting in an increase in ATP synthesis capacity.

During hyperthyroidism, there is an increase in energy dissipation in the form of heat (uncoupling), while during hypothyroidism, there is a decrease in the maximum rate of ATP synthesis.

## Conclusion

In conclusion, the respiratory chain and oxidative phosphorylation constitute a highly efficient, organized system supported by the inner membrane of the mitochondrion. This orderly assembly of proteins plays a key role in converting the energy released during substrate oxidation into chemical energy stored in the form of ATP. This energy molecule is then made available to cells to fuel various biosynthesis reactions and maintain their vital functions.